

Maurice D. Hanisch

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Education

California Institute of Technology

Ph.D. in Computing and Mathematical Sciences, GPA: 3.94

Pasadena, USA

09/2025 – 06/2029

- Advised by Prof. **Anima Anandkumar**; fully funded by the **Kortschak Fellowship**.

ETH Zurich

M.Sc. in Physics, GPA: 5.77/6.00 (Top 12%)

Zurich, Switzerland

09/2022 – 05/2024

Ludwig Maximilian University of Munich

B.Sc. in Physics, GPA: 1.28/1.00 (Top 10%)

Munich, Germany

10/2018 – 09/2022

Experience

California Institute of Technology

Graduate Researcher, A. Anandkumar's Group

Pasadena, USA

09/2025 – Present

- Designed and developed (co-first author) an SE(3)-equivariant **Fourier neural operator** that learns the Kohn-Sham map, replacing diagonalizations in **density functional theory (DFT)** for quasi-linear scaling (**runtime exponent 1.03 vs. 3.37** for Quantum ESPRESSO).
- Built the training stack in **PyTorch Lightning** (**multi-GPU DDP**, Slurm, L40S/H200/B300) and scaled to **8,250 atoms on one B300 GPU**.
- Modified **Quantum ESPRESSO** (Fortran/MPI) and built the full data pipeline, generating **59,133 potential-density pairs** (8,504 structures).

Bloomberg CTO Office

Security Research Intern, J. Vanegue's Team

New York, USA

06/2025 – 09/2025

- Found attack vectors against the **CHERI** secure CPU architecture; co-author of an **ACM CCS 2026** paper, a tier-1 security venue.

California Institute of Technology

Machine Learning Research Intern, A. Anandkumar's Group

Pasadena, USA

09/2024 – 05/2025

- Led a benchmark **retraining 7 state-of-the-art molecular models** (e.g., EquiformerV2) on **charged and open-shell** molecules.
- Built a differentiable **xTB** (tblite) pipeline feeding **orbital-based quantum features** to **OrbitAll's** SE(3)-equivariant GNNs.

IBM Research Europe

Master's Thesis, S. Woerner's Group

Zurich, Switzerland

09/2023 – 05/2024

- Built a Python/C++ soft-decoding pipeline processing **billions of measurement records** for one of IBM's then-largest QEC experiments.
- Ran the full experimental pipeline in **Qiskit** on a 127-qubit IBM Quantum device (error-correcting repetition codes up to distance 51), showing that soft (analog) readout decoding raises the **repetition-code threshold by 24%** and cuts logical error rates by up to **30×**.

ETH Zurich

Summer Project, J. Home's Group

Zurich, Switzerland

06/2023 – 08/2023

- Engineered and **experimentally validated** ion-trap potentials for a tunable interaction between **GKP qubits** on trapped-ion hardware.

Max Planck Institute of Quantum Optics

Bachelor's Thesis, I. Cirac's Group

Garching, Germany

04/2022 – 09/2022

- Proved that **bosonic Gaussian quantum circuits** can be implemented with **exponentially fewer qubits** than a direct encoding.

Selected Publications

Learning the Kohn-Sham map with neural operators for quasi-linear scaling density functional theory

D. Khan*, **M.D. Hanisch***, N. Argatoff, E. Xie, S. Sharma, A. Anandkumar. Under review at **Science**

Soft information decoding with superconducting qubits

M.D. Hanisch, B. Hetényi, J.R. Wootton. **APS Open Sci.** 1, 000019 (2026)

OrbitAll: A Unified Quantum Mechanical Framework for Deep Learning Across Diverse Molecular Regimes

B.S. Kang*, V.C. Bhethanabotla*, A. Tavakoli, **M.D. Hanisch**, ..., A. Anandkumar. Under review at **Nature Computational Science**

Awards and Competitions

- Joined 10+ competitions and workshops (NVIDIA, Bloomberg, EDF); won over \$3,500 in hackathon prizes (RNNs, LLMs, CUDA-Q).

Hackathons: **1st/30** ETH QHack 2024 (NVIDIA QAOA task) | **1st/12** LOQC 2023 (EDF task) | **2nd/6** ETH QHack 2023 (IQM task)

Skills

ML/deep learning: PyTorch (Lightning), neural operators (FNO), equivariant GNNs (e3nn), transformers, LLMs, multi-GPU DDP

Scientific computing: DFT/electronic structure (Quantum ESPRESSO, PySCF), xTB/tblite, ASE, RDKit, numerical linear algebra

Quantum computing: Qiskit, quantum error correction, Hamiltonian simulation

Programming: Python, C++, C, Fortran, Linux, Slurm/HPC, Git, Docker

Open source: Merged PRs to neuraloperator (FNO), tblite (xTB), HyperQueue (HPC), cheribsd

Languages: German (native), French (native), English (C1: IELTS 8/9), Italian (A1)

Interests: Weightlifting, volleyball, calisthenics, surfing, chess